# BRD: Omnichannel Order & Inventory Platform

Business Requirement Document - Enterprise Spec v2.4

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## 1.0 Business Context and Executive Summary

This project establishes a unified Omnichannel Order Orchestration and Real-time Inventory Reconciliation Engine across e-commerce storefronts, mobile channels, and ERP. Expected outcome: 40% reduction in fulfillment latency, zero inventory drift, 99.98% sync SLA.

## 2.0 Stakeholders and RACI Matrix

Product Sponsor: VP of Retail Operations (Accountable)
Technical BA: Vibhor (Responsible for requirements, specs, BDD criteria, and system flows)
Engineering Lead: Chief Architect (Consulted on API contracts and microservice latency limits)
QA and Compliance: Security & Regulatory Officer (Informed on tokenized payment PCI-DSS)

## 3.0 Functional Scope and User Stories

FR-101 [Soft Reservation]: When a customer reaches step 2 of checkout, reserve SKU for 15 min.
FR-102 [Distributed Concurrency]: Prevent overselling when concurrent checkouts exceed stock count.
FR-103 [Payment Webhook Idempotency]: Handle duplicate payment notifications without double-charging.
FR-104 [ERP Synchronization]: Push confirmed orders to SAP within 500ms via Kafka topic.

# BRD: Omnichannel Order & Inventory Platform - Detailed Specifications

## 4.0 Non-Functional Requirements (NFRs)

NFR-01 Performance: P99 latency for stock reservation under 120ms during peak 50,000 RPM load.
NFR-02 Availability: 99.99% high-availability multi-region active-active deployment on AWS.
NFR-03 Security: Full end-to-end TLS 1.3 encryption, OAuth 2.0 JWT bearer tokens, and PCI-DSS compliance.

## 5.0 BDD Acceptance Criteria (Given-When-Then)

Scenario: Concurrent purchase of final inventory unit
  Given SKU-1048 has an available stock quantity of 1
  When User A and User B trigger checkout confirmation within 50ms
  Then User A acquires the distributed Redis lock and receives Order Confirmation
  And User B receives an out-of-stock notification with an alternative recommendation.